GeoGebra – 7th Standard, Term 3

Algebra

Geometric Representation of the identity

$$(a + b)^2 = a^2 + 2ab + b^2$$

Question:

Evaluate 15^2 using the identity

$$(a + b)^2 = a^2 + 2ab + b^2$$

1. Open GeoGebra

Open GeoGebra.

Choose Calculator Suite.

2. Create two sliders named a and b

These sliders allow us to change the values of a and b dynamically.

Whenever we change a slider, the dimensions and areas change automatically. This helps us observe how algebraic identity works for different values.

Slider settings:

```
Name: a | Min: 1 | Max: 10 | Step: 1
```

```
Name: b | Min: 1 | Max: 10 | Step: 1
```

3. Create four regions

In the Input Bar, enter the following commands:

- Region 1

```
P1=Polygon((0,0),(a,0),(a,a),(0,a))
```

Area: a^2

- Region 2

```
P2=Polygon((a,0),(a+b,0),(a+b,a),(a,a))
```

Area: ab

- Region 3

```
P3=Polygon((0,a),(a,a),(a,a+b),(0,a+b))
```

Area: ab

- Region 4

```
P4=Polygon((a,a),(a+b,a),(a+b,a+b),(a,a+b))
```

Area: b^2

4. Add the area labels

In the Input Bar, enter the following commands one by one:

```
Text("a²",(a/2,a/2))
```

```
Text("ab",(a+b/2,a/2))
```

```
Text("ab",(a/2,a+b/2))
```

```
Text("b²",(a+b/2,a+b/2))
```

These commands place the area labels at the approximate centres of the four regions.

5. Display the identity

In the Input Bar, enter:

```
Text("(a+b)² = a² + 2ab + b²")
```

```
Text("(a+b)² = a² + ab + ab + b²")
```

6. Display the values of area

In the Input Bar, enter the following commands:

```
Text("a² = " + a^2)
```

```
Text("ab = " + a*b)
```

```
Text("ab = " + a*b)
```

```
Text("b² = " + b^2)
```

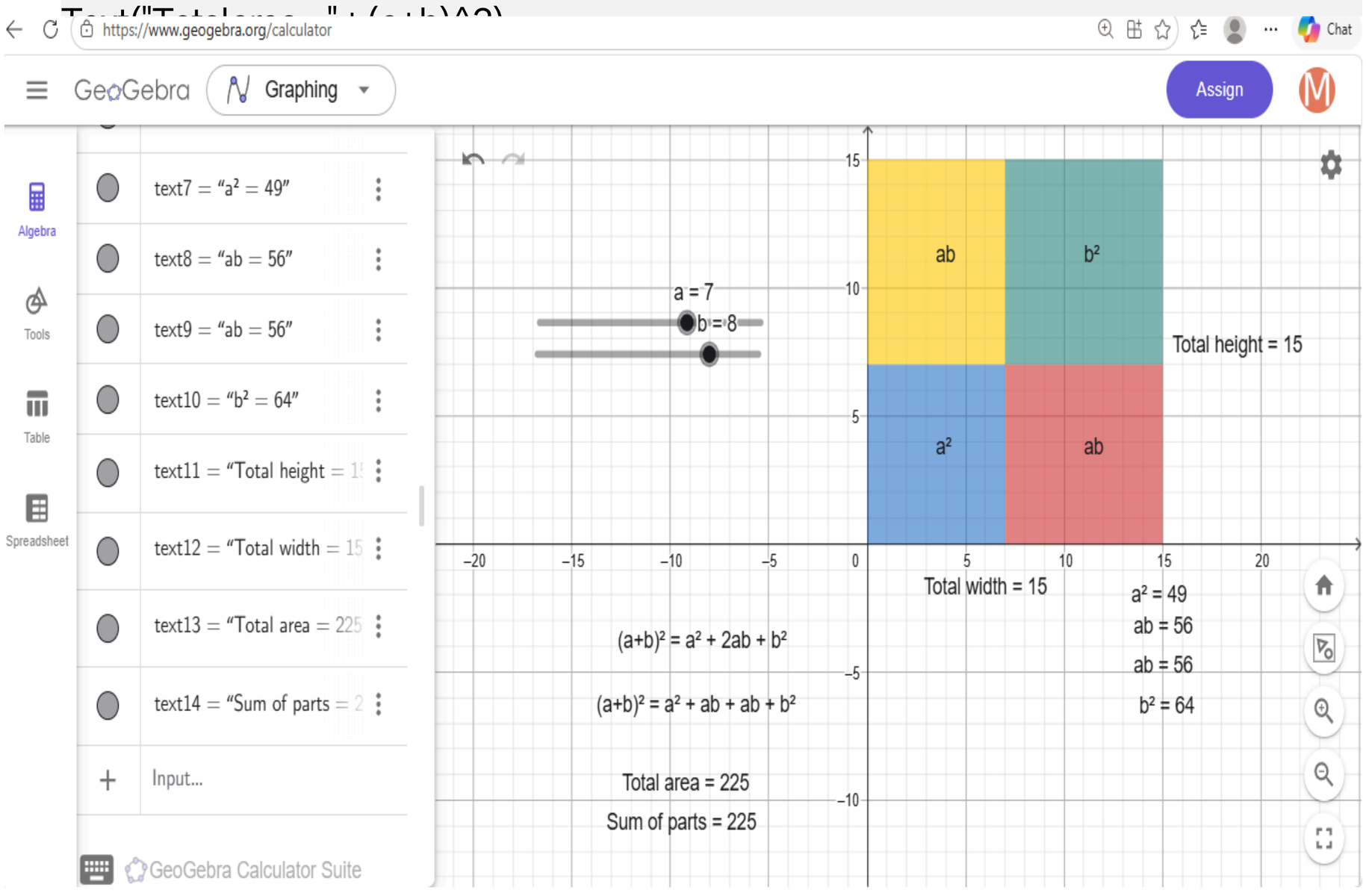
7. Display total height, total width, and area

In the Input Bar, enter the following commands (add a suitable position coordinate to each Text command if GeoGebra requires it):

```
Text("Total height = " + (a+b), (a+b+0.5, (a+b)/2))
```

```
Text("Total width = " + (a+b))
```

```
Text("Total area = " + (a+b)^2)
```



Factorisation Using Identities

To help students factorise an expression of the form $z^2 - n$ using the identity $a^2 - b^2 = (a + b)(a - b)$, where n is a perfect square.

Question:

Factorise: $z^2 - 49$.

GeoGebra setup Question:

1. Open GeoGebra and choose Calculator Suite.

2. Use the Input Bar to enter the commands below, one at a time.

Step 1 – Set the value of n

```
n=49
```

You may change 49 to another positive perfect-square number using Inputbox.

Step 2 – Display the question

```
question=Text("Factorise: z^2 − "+n)
```

Step 3 – Calculate the correct value

```
correctB=sqrt(n)
```

The expected value of b is $\sqrt{n}$. This activity assumes n is a perfect square.

Step 4 – Create an Input box

1. Create the number b first (for example, b = 0), then use the Input Box tool and link it to b.
2. Open Tools and select Input Box.

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box in the Graphics view.

) and Link Object = b

Step 5 – Check the answer

```
check = (b == correctB)
```

Step 6 – Display feedback

```
feedback = If(check, "Correct!! Well done!!", "Try again!")
```

The feedback changes automatically when b changes.

Step 7 – Show the identity

```
identity = Text("Identity: a² − b² = (a + b)(a − b)")
```

Use this identity to explain the factorisation.

Step 8 – Show the substitution

```
substitution = Text("Since n = " + n + ", b = " + correctB)
```

Step 9 – Show the difference of squares

```
difference = Text("z² − " + n)
```

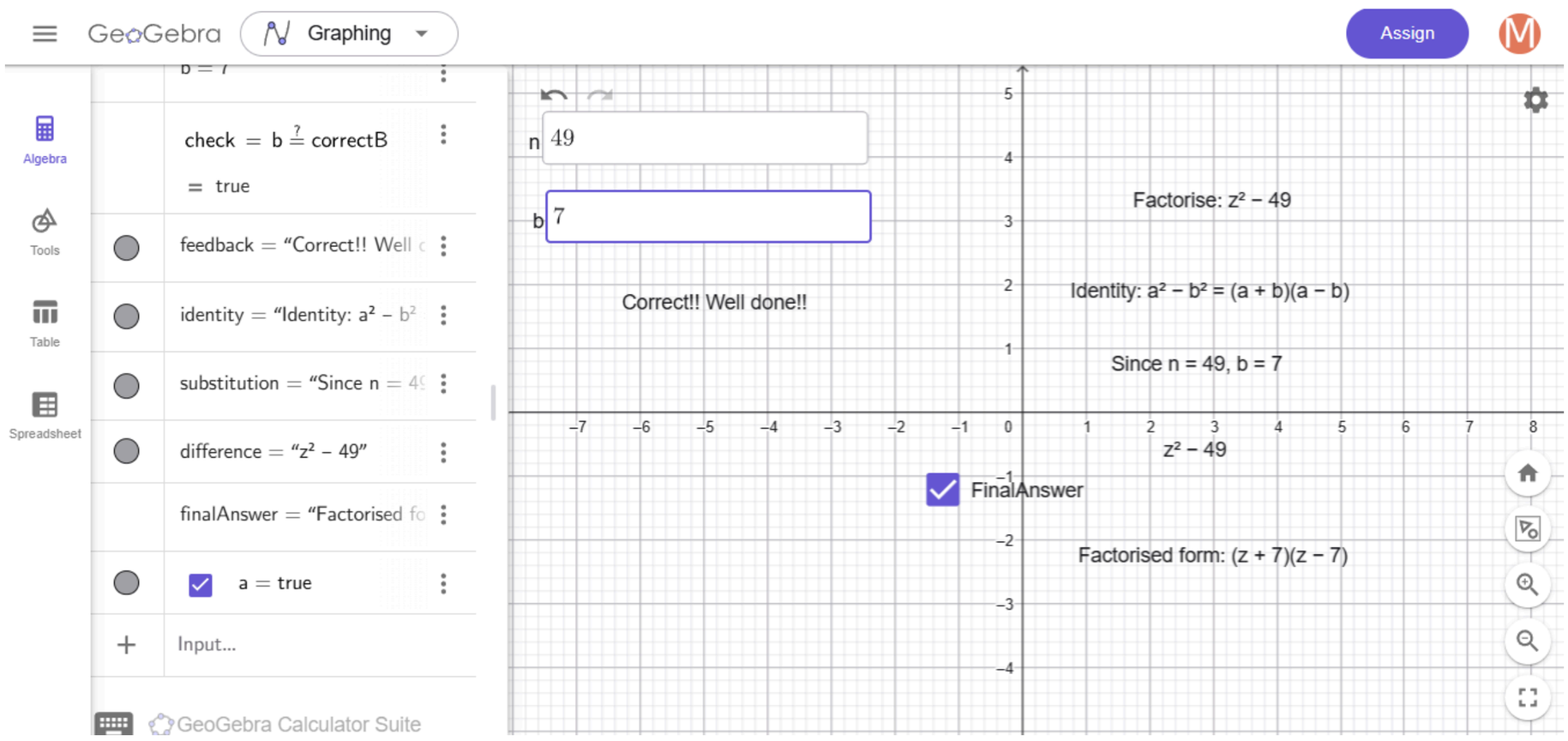
This repeats the expression in the question.

Step 10 – Show the final factorised form

```
finalAnswer = Text("Factorised form: (z + " + correctB + ")(z − " + correctB + ")")
```

For n = 49, the result is (z + 7)(z − 7).

Output:



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Graphical representation of Inequation

Two forms of inequalities:

1. Addition/Subtraction form: $a<b+x<c$

2. Multiplication/Division form: $a<bx<c$

Question:

Solve the inequation $-2<x+3<5$, where x is an integer, and represent the solution graphically.

Open GeoGebra and choose Calculator Suite.

Step 1: Enter the three integers

```
a=-2
```

```
b=3
```

```
c = 5
```

For multiplication, keep b positive and non-zero.

Step 2: Choose the operator

```
isAdd = true
```

This Boolean variable selects the addition form when it is true. To switch to multiplication, change it to false.

Step 3: Display the question

```
text1 = Text(If(isAdd, "Solve: " + a + "<" + b + "x<" + c, "Solve: " + a + "<" + b + "x<" + c))
```

Step 4: Explain the inverse operation

```
text2 = Text(If(isAdd, "Subtract " + b + " from all three parts", "Divide all three parts by " + b
```

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Step 5: Display the transformed inequality

```
text3 = Text(If(isAdd, "a - b = " + (a - b) + "<x<c−  b = " + (c - b), "a/b = " + (a/b) + "<x<c/b = " + (c/b)))
```

This step displays the transformed inequality after applying the selected inverse operation.

Step 6: Calculate the integer limits

```
xmin = If(isAdd, a - b + 1, floor(a/b) + 1)
```

```
xmax = If(isAdd, c - b - 1, ceil(c/b) - 1)
```

These limits are based on strict inequalities. If you change the inequality signs, adjust the limits accordingly.

Step 7: Create the integer solution points

```
L = Sequence((n, 0), n, xmin, xmax)
```

Step 8: Display the changing solution set

```
Text("Smallest integer solution = " + xmin)
```

```
Text("Largest integer solution = " + xmax)
```

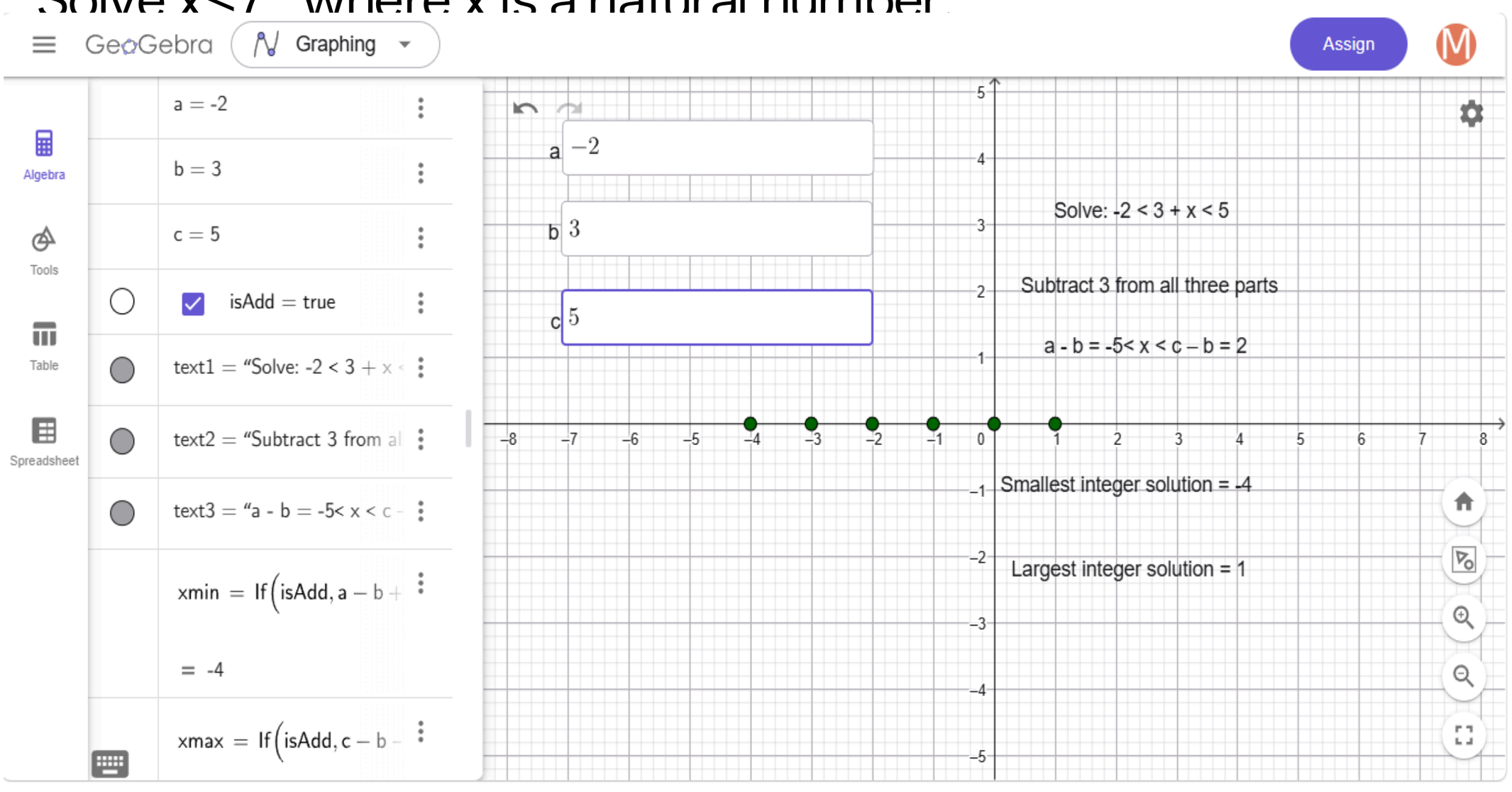
Change a, b, c, or isAdd and observe the transformed inequality and integer solution points.

Output:

Simple Linear Inequation on a Number Line

Question:

Solve x<7 where x is a natural number



```
isWhole = false
```

```
isInteger = false
```

Step 4: Select the inequality

```
IncludeN = true
```

IncludeN = true means x ≤ n. IncludeN = false means x<n.

Step 5: Display the question

```
questionText = Text(If(IncludeN, "Solve: x ≤ " + n, "Solve: x < " + n))
```

Step 6: Create the lower limit

```
m = -10
```

For the integer demonstration, create an Input Box linked to m so that the visible range ca be changed.

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Step 7: Calculate and display the solution limits

(i) Calculate the first displayed solution:

```
start = If(isNatural, 1, If(isWhole, 0, m))
```

(ii) Calculate the last solution:

```
last = If(IncludeN, n, n - 1)
```

Step 8: Generate the answer list

```
answerList = Sequence(k, k, start, last)
```

Step 9: Display the answer list

```
answerText = Text("Solutions: " + answerList)
```

Step 10: Display the number line

```
Sol = Sequence((k, 0), k, start, last)
```

How to use the demonstration

- Change n using the Input Box to update the inequality.
- Set IncludeN to true for x ≤ n or false for x<n.
- Choose natural, whole, or integer values using the Boolean variables.
- For integers, change m to control the first displayed value.
- Observe the answer list and number-line points update.

Note: Use only one of isNatural, isWhole, isInteger as true at a time.

Output:

Proving Algebraic Difference Identities

Question:

Show that $(x + 2y)^2 - (x - 2y)^2 = 8xy.$

Open GeoGebra and choose Calculator Suite.

Step 1: Create a Slider

```
a=2

Minimum:1

Maximum:50

Step: 1
```

Step 2: Display the question

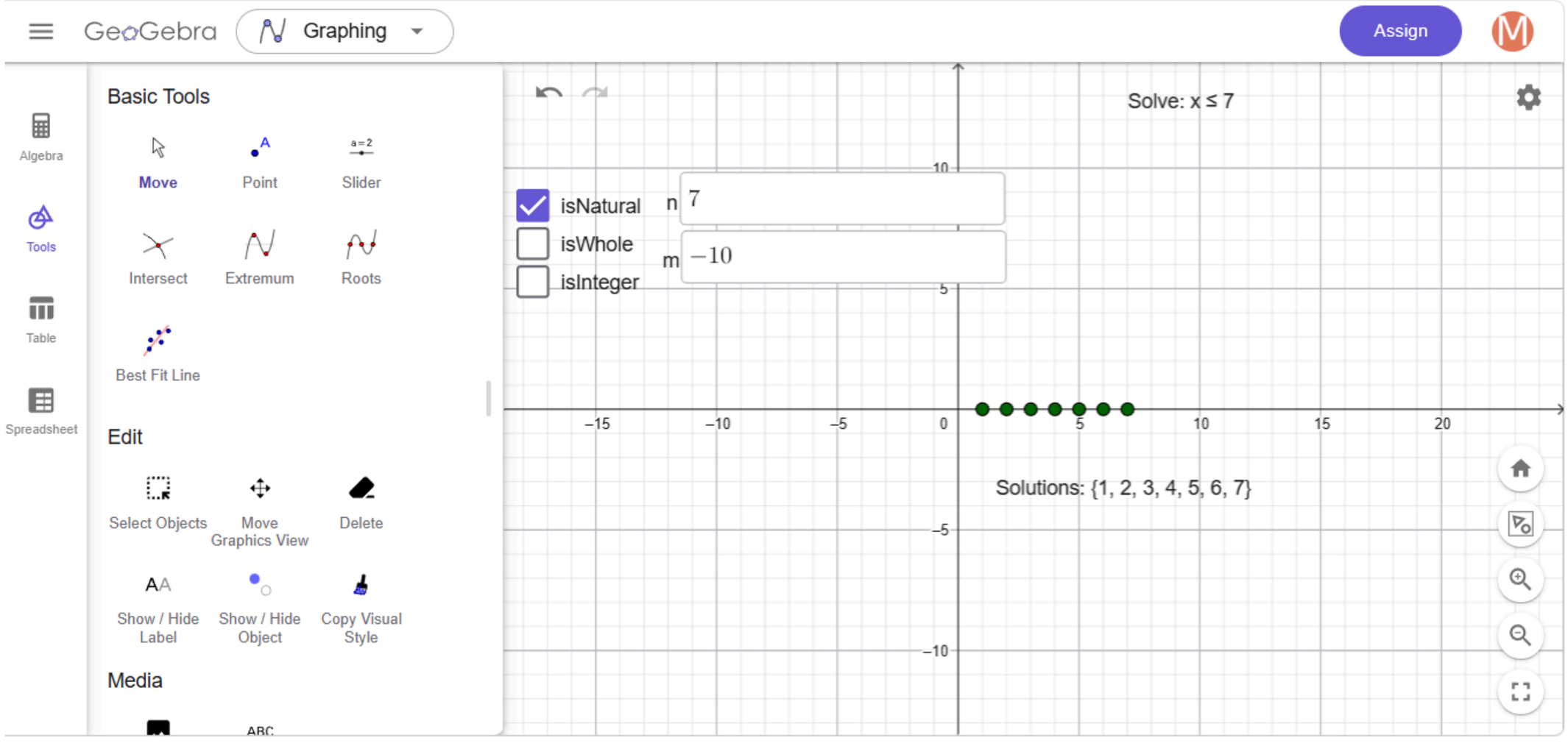
```
Text("Question:(x+"+a+"y)^2-(x-"+a+"y)^2")
```

Step 3: Display the expansion of the first square

```
Text("(x+"+a+"y)^2=x^2+"+(2*a)+"xy+"+(a^2)+"y^2")
```

Step 4: Display the expansion of the second square

```
Text("(x-"+a+"y)^2=x^2-"+(2*a)+"xy+"+(a^2)+"y^2")
```



Step 5: Display the subtraction

```
Text("x^2+"+(2*a)+"xy+"+(a^2)+"y^2-x^2+"+(2*a)+"xy-"+(a^2)+"y^2")
```

Step 6: Display the final result

